

The Editorial Office, *Quantum Reports* (MDPI)

Re: Submission of “Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus”

Dear Editors,

We are pleased to submit our manuscript, “*Predicting the Trainability of Variational Quantum Circuits: A Data-Driven Model for Barren Plateaus*,” for consideration as an original research article in *Quantum Reports*.

Significance. Barren plateaus—the exponential vanishing of cost-function gradients as a variational quantum circuit is scaled—are widely regarded as the central obstacle to training quantum machine-learning models. Diagnosing them today requires expensive gradient sampling on each candidate circuit, infeasible precisely in the large-qubit regime where the problem matters most. Our manuscript reframes this diagnosis as a *supervised prediction* problem: can a purely classical model predict a circuit’s trainability from its architecture alone—qubit count, depth, ansatz type, entanglement pattern, entangler gate, and cost-observable locality—without evaluating a single gradient? Using an exact statevector simulator we generate a labeled dataset of 20,000 random circuits and show that a gradient-boosted model predicts the gradient-variance trainability with high accuracy (AUC 0.99 for the barren/trainable decision) and, most notably, *extrapolates* from small circuits to the largest sizes we can still simulate—yielding a cheap ansatz-screening heuristic and an interpretable account of which design choices drive untrainability.

Context and relationship to existing work. The barren-plateau literature is overwhelmingly analytical, deriving variance scaling laws for particular ansatz families (e.g., McClean et al. 2018; Cerezo et al. 2021; Larocca et al. 2025). Our contribution is complementary and empirical: we learn a predictive map across a heterogeneous design space and show that it recovers the established theory from data. Consistent with recent skeptical benchmarking (Bowles et al. 2024; Schuld & Killoran 2022), we make **no** claim of quantum advantage—the object we predict is a property *of* quantum circuits, while the predictor is a classical diagnostic tool.

Fit to the scope of Quantum Reports. The manuscript sits squarely within the journal’s remit of quantum computing, quantum information, and quantum machine learning, and is a natural fit for the Special Issue “*The Integration of Quantum Computing with Artificial Intelligence*.”

Reproducibility. In keeping with the journal’s open-science standards, all code, fixed random seeds, and the full dataset are openly available at https://github.com/bithabib/quantum_computer_simulator, and the underlying simulator runs interactively in the browser at <https://quantum.qbithabib.com>; every reported number regenerates from a single command.

Required statements.

- We confirm that neither the manuscript nor any parts of its content are currently under consideration for publication with or published in another journal.
- All authors have approved the manuscript and agree with its submission to *Quantum Reports*.
- This manuscript has not been previously submitted to any MDPI journal.
- The authors declare no conflicts of interest.

We believe this work will interest the readership of *Quantum Reports* and look forward to the reviewers’ feedback.

Sincerely,

Jaeho Kim (on behalf of all authors)

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